

TABLE II
AVR910 UART Serial Commanding Protocol

	Host PC writes		Host PC reads	
	ID	Data	ID	Data
Enter programming mode	'P'	—	—	13d
Report auto increment address	'a'	—	—	'Y'
Set address	'A'	ah al	—	13d
Write program memory, low byte	'c'	Dd	—	13d
Write program memory, high byte	'C'	Dd	—	13d
Issue page write	'm'	—	—	13d
Read program memory	'R'	—	dd(dd)	—
Write data memory	'D'	dd	—	13d
Read data memory	'd'	—	dd	—
Chip erase	'e'	—	—	13d
Write lock bits	'l'	dd	—	13d
Write fuse bits	'f'	dd	—	13d
# Read fuse and lock bits	'F'	—	dd	—
Leave programming mode	'L'	—	—	13d
Select device type	'T'	dd	—	13d
Read signature bytes	's'	—	3*dd	—
Return supported device codes	't'	—	n*dd	00d
Return software identifier	'S'	—	s[7]	—
Return software version	'V'	—	dd dd	—
Return hardware version	'v'	—	dd dd	—
Return programmer type	'p'	—	dd	—
Set LED	'x'	dd	—	13d
Clear LED	'y'	dd	—	13d
Universal command	'.'	3*dd	dd	13d
New universal command	'.'	4*dd	dd	13d
Special test command	'Z'	2*dd	dd	—

tomatically erased before new data is written. If polling is not used, the user must wait for a period of at least t_{WD_EEPROM} before issuing the next byte. In a chip-erased device, no 0xFFs in the data files need to be programmed.

6. Any memory location can be verified by using the read instruction, which returns the content at the selected address at serial output MISO.

7. At the end of the programming session, RST can be set high to commence normal operation.

8. Power-off (if needed): Set RST to '1' and turn V_{CC} power off.

Programming the target AVR microcontroller (for user interface) using Visual Basic .NET (VB.NET). Once the master device (IC2) is ready to receive data or command from the PC and is connected to the slave device (IC5), it only requires an interface software on the PC which can supply data/com-

mand to it. This interface software is written in Visual Basic .NET. It supports AVR910 in-system programming protocol for commands. The interface software (ADU-ISP) given here can implement almost all ATmega AVR microcontrollers having Flash size of less than 128 kB. Just construct a form in VB which has the controls as shown in Fig. 6, with SerialPort and OpenFileDialog controls, and insert the code that follows.

The source program includes remarks for easy understanding.

Table II shows AVR910 UART serial commanding protocol. These commands are sent to the program-

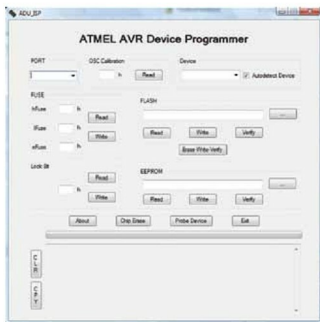


Fig. 6: VB.NET form layout for programmer software

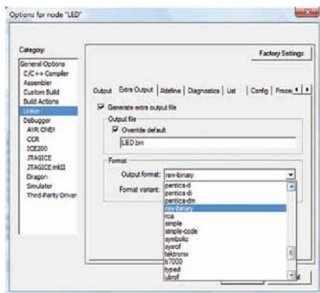


Fig. 7: Screenshot of raw binary file generation

mer board through serial communication using the VB.NET program. All commands comprise a single letter. The programmer returns 13d, which is a carriage-return character or the data read after the command is finished.

The VB.NET program accepts raw binary file (*.bin) file, which is generated after compiling and linking a source program for the microcontroller. This binary file data has to be burnt into the